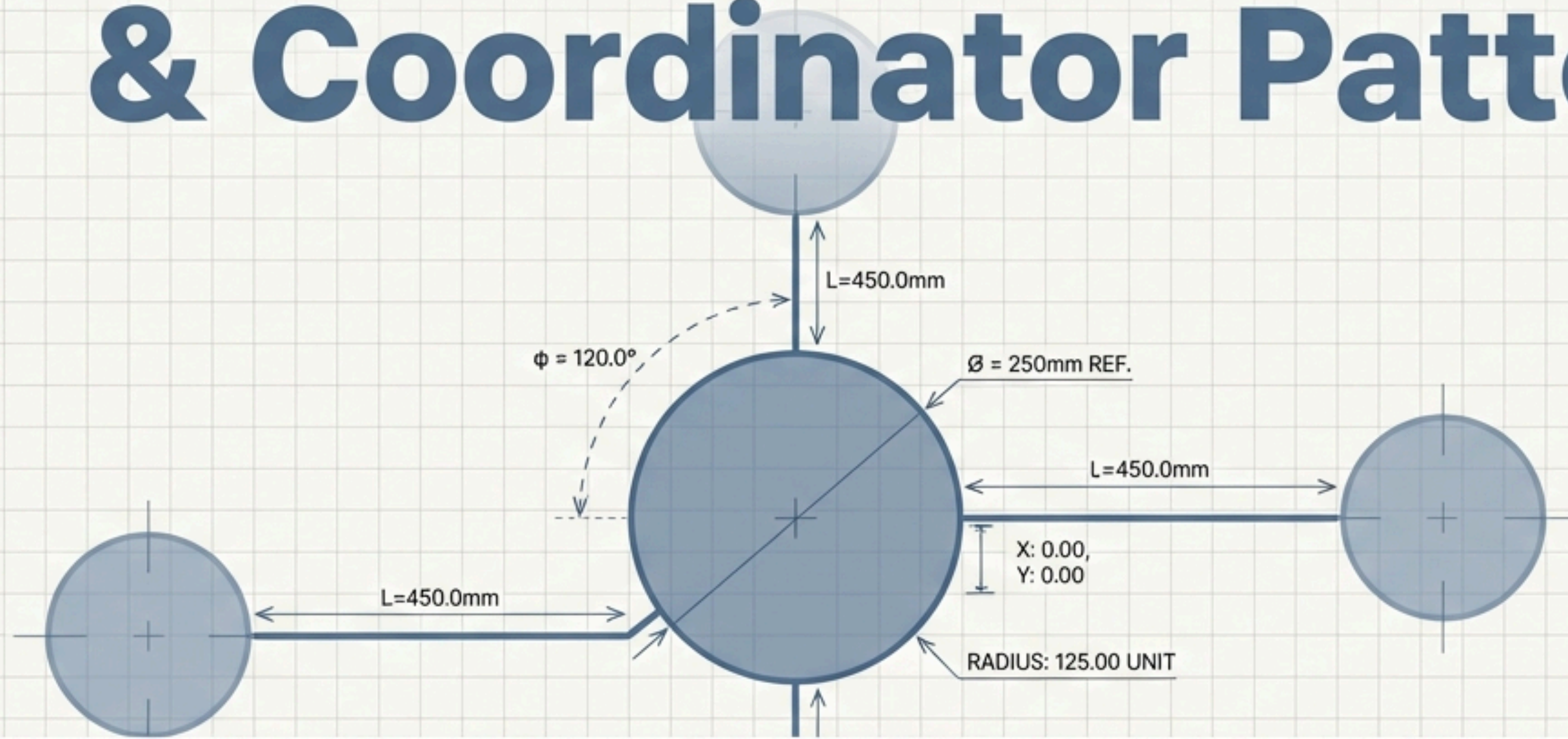


EXAM ALERT:
DOMAIN 1

Multi-Agent Systems & Coordinator Patterns



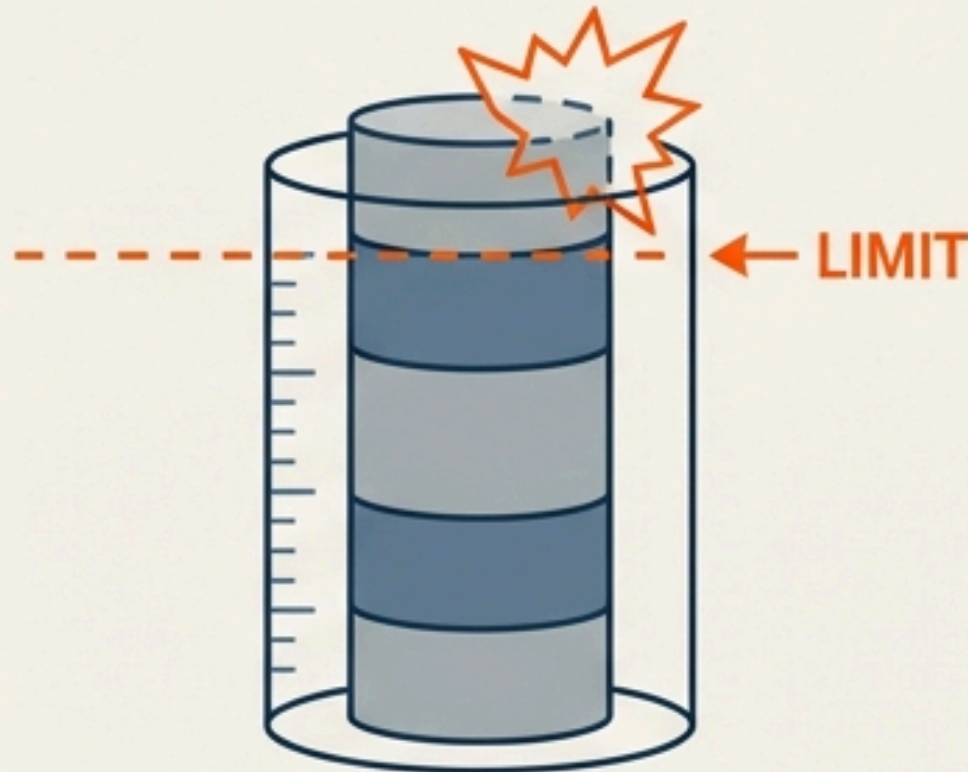
Module: Claude Certified Architect Course | Module 1

Domain: Domain 1 — Agentic Architecture & Orchestration (27% of exam)

Task Statements: 1.2 (Orchestrate multi-agent systems) & 1.3 (Configure subagent invocation)

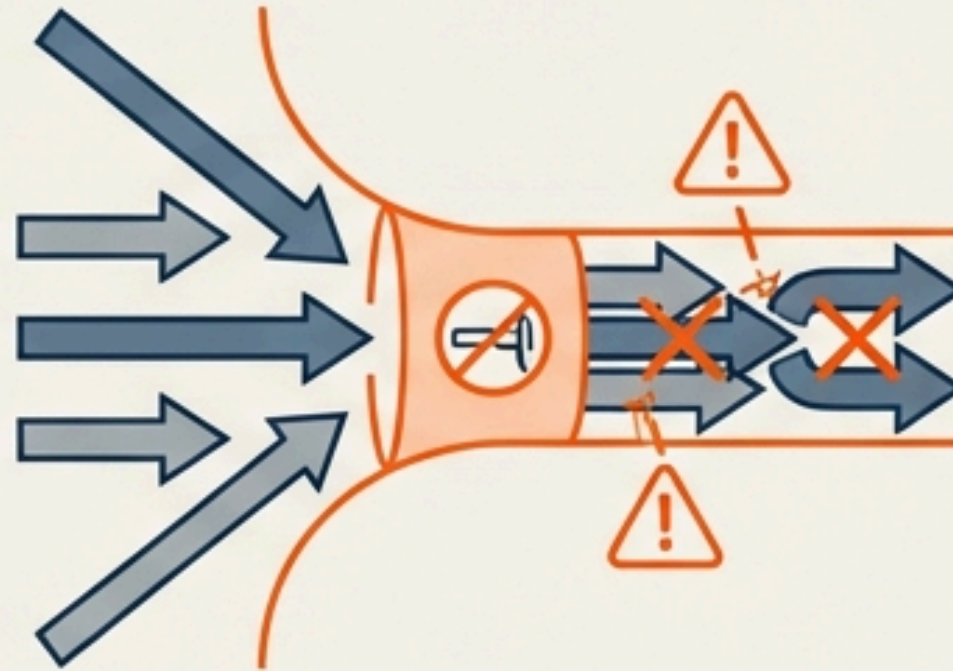
System Warning: The Single-Agent Ceiling

Context Overflow



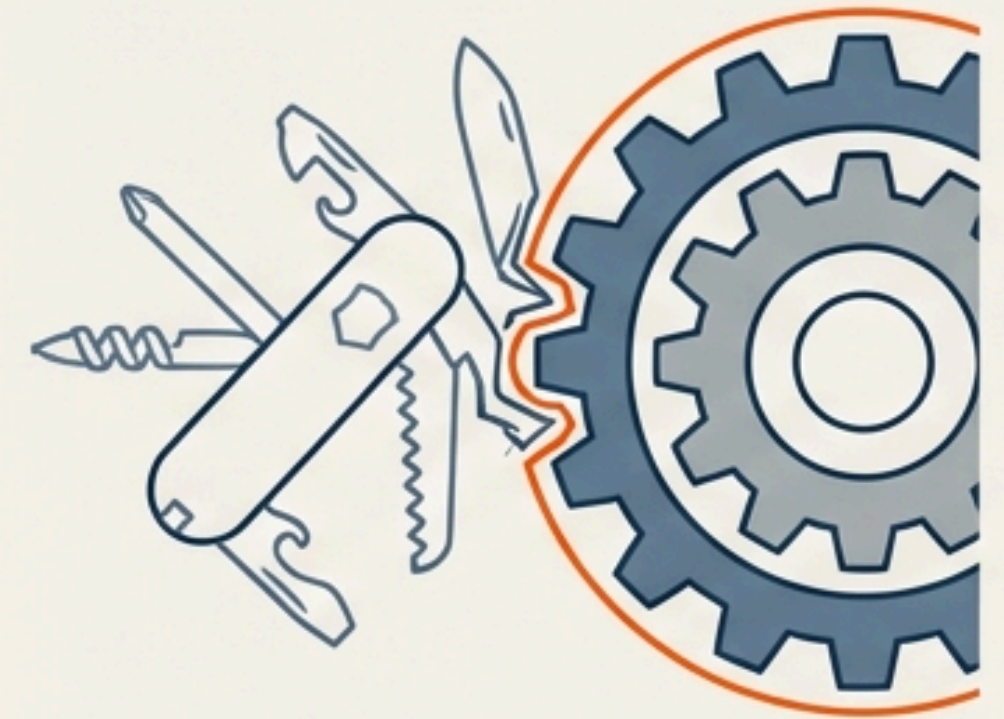
Tool results accumulate rapidly. Complex tasks hit the context ceiling before completion.

Sequential Bottlenecks



Single agents are strictly sequential. Independent subtasks queue up, compounding system latency.

Specialization Gap

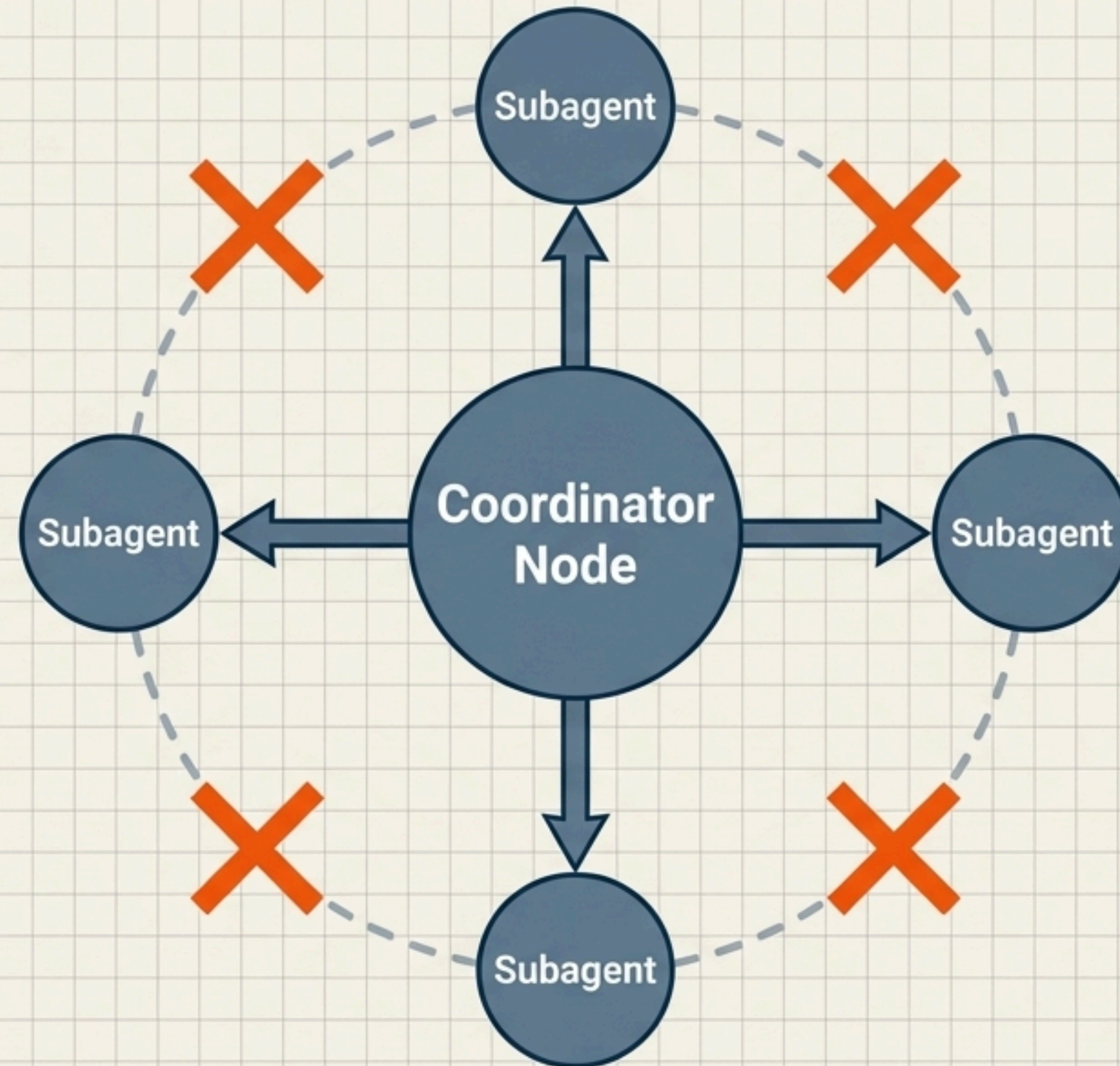


Generic instructions yield generic results. Scoped agents with narrow jobs perform measurably better.

The Blueprint Fix: Multi-agent architectures deploy specialized agents in parallel, each with a scoped context.

The Hub-and-Spoke Topology

The Core Rule:
Subagents NEVER
communicate directly
with each other.

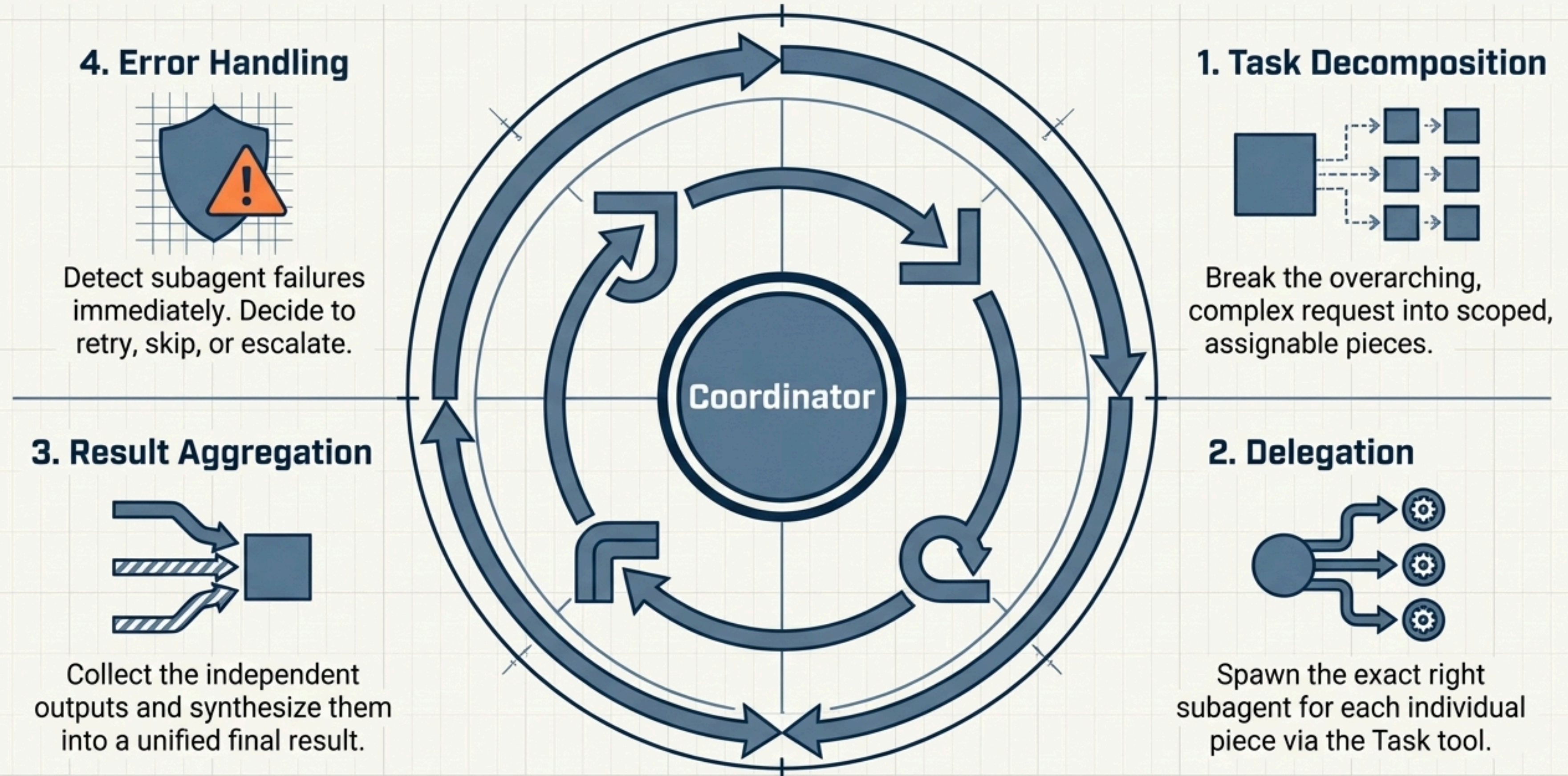


The Routing Mandate:
Absolutely all
communication routes
exclusively through the
central coordinator.

Exam Tip

This topology is the only
way to guarantee
complete system
observability and
centralized error handling.

The Four Coordinator Responsibilities

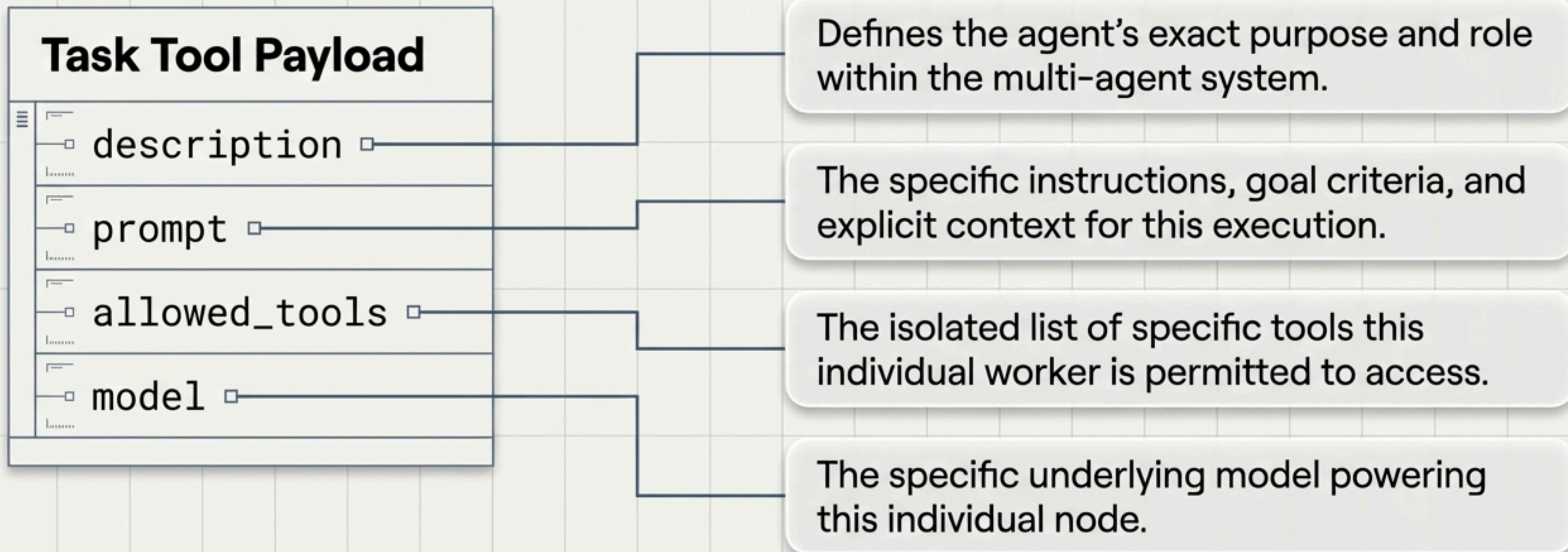


Task Tool Permission Matrix

Configuration rules for allowedTools

Agent Role	Task Tool Access	Architectural Rationale
Coordinator	 YES	Absolutely required. Without the Task tool, the coordinator cannot spawn any subagents.
Standard Subagent	 NO	Standard workers must not spawn further agents. Uncontrolled recursive delegation breaks the system.
Hierarchical Sub-Coordinator	 YES	Allowed only if this specific node is explicitly designed to delegate to a deeper, tertiary layer.

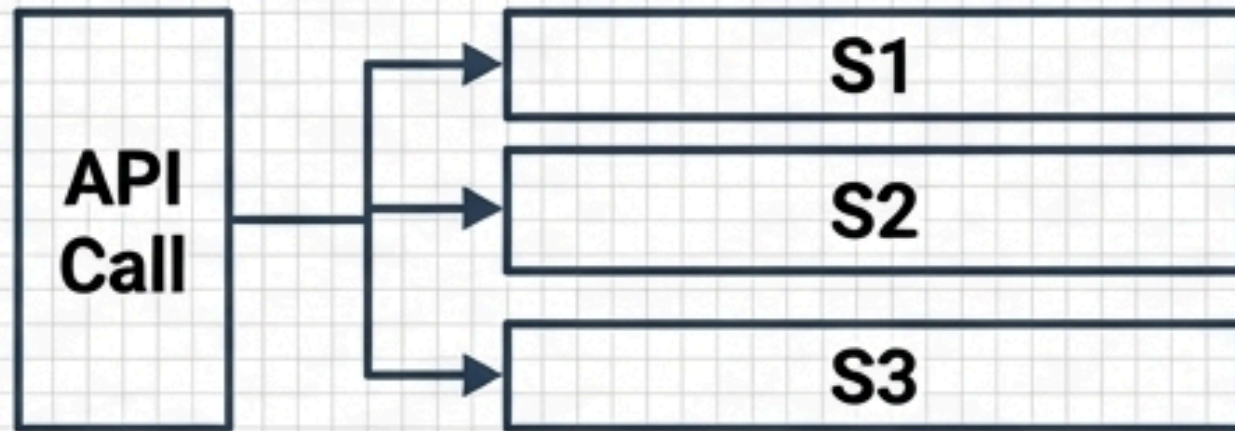
Blueprinting the AgentDefinition



The Mechanism: When a coordinator calls the Task tool, it must pass a strict AgentDefinition object to successfully spawn the subagent.

Execution Latency: Parallel vs. Sequential Spawning

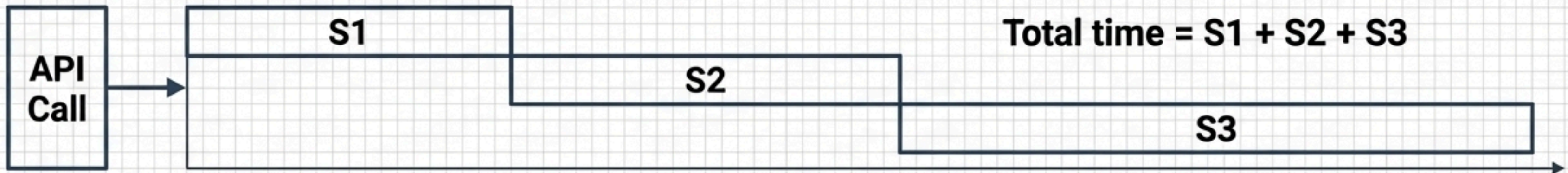
✓ Parallel Execution - Correct



Total time = $\max(S1, S2, S3)$

Rule: Emit multiple Task tool calls inside a single coordinator response.

✗ Sequential Execution - Anti-Pattern



Total time = $S1 + S2 + S3$

Exam Diagnostic Alert

If a system runs 3x slower than expected, the coordinator is committing the sequential anti-pattern (waiting turn-by-turn).

The Context Isolation Airlock

Passing Through

System prompt

Task tool prompt field

allowed_tools list

model setting



Blank Slate
Subagent

Blocked

Coordinator's conversation
history (turns 1-N)



Prior tool results

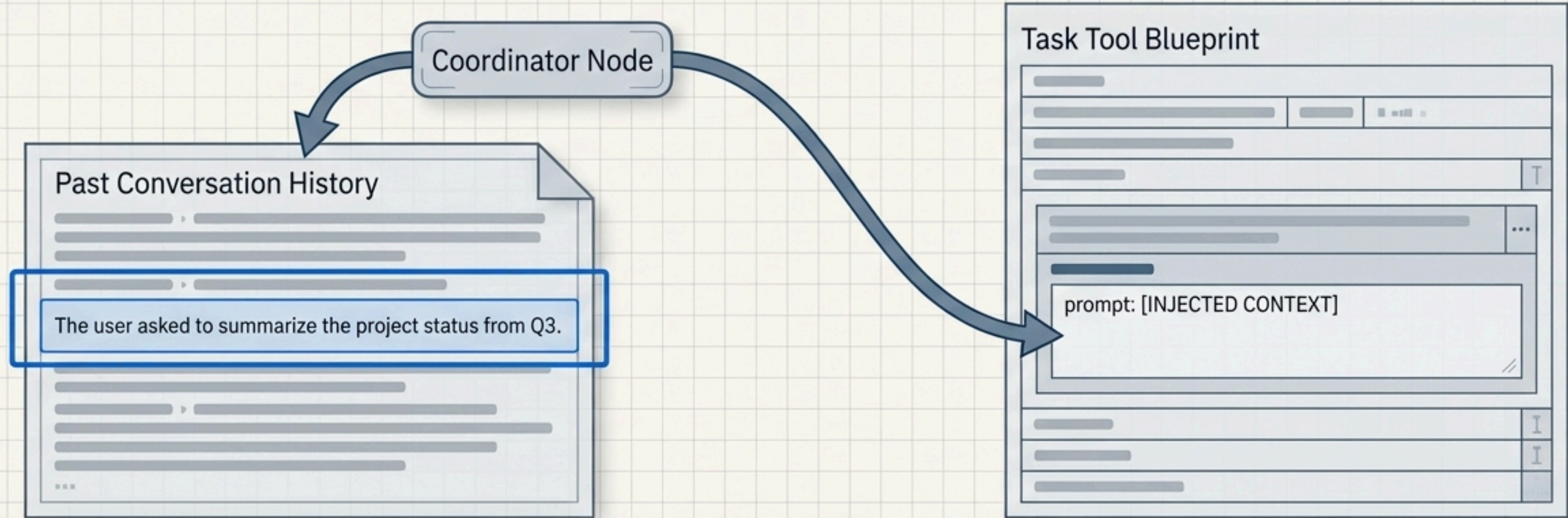


Inferred/ambient context



The **#1 Exam Gotcha (Task 1.3)**: Subagents start with a **completely empty brain**. **Zero historical context** automatically crosses the barrier. They inherit **absolutely nothing**.

Bypassing the Airlock: Explicit Injection



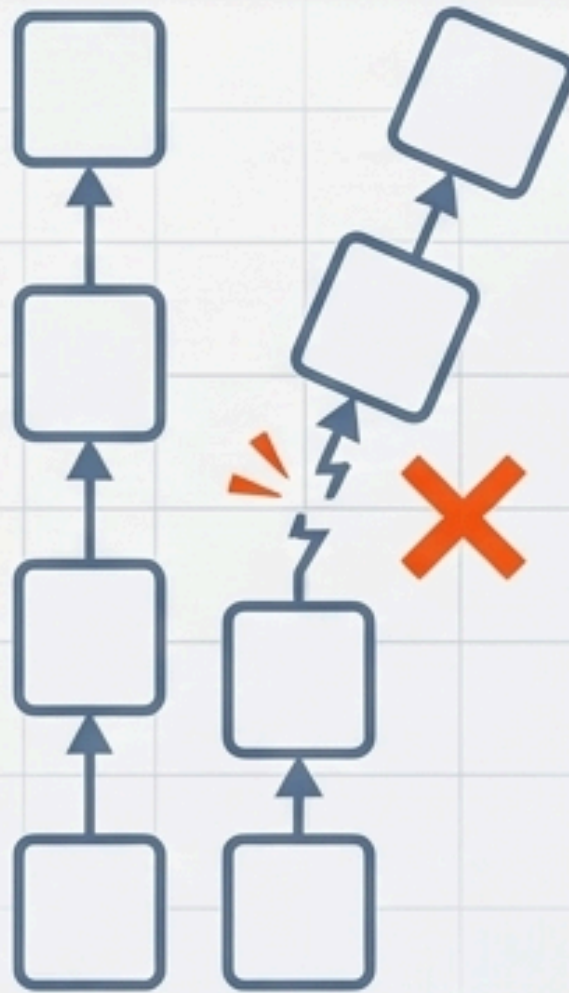
- **The Fix:** If a subagent needs historical information to do its job, the coordinator must actively extract it.
- **Action Required:** Explicitly inject the required data into the prompt field of the Task tool call.

Diagnostic Alert: If a subagent produces **irrelevant, duplicated, or context-unaware** output on the exam, the root cause is missing **explicit context injection**.

Orchestration Prompts: Goals over Steps

Anti-pattern

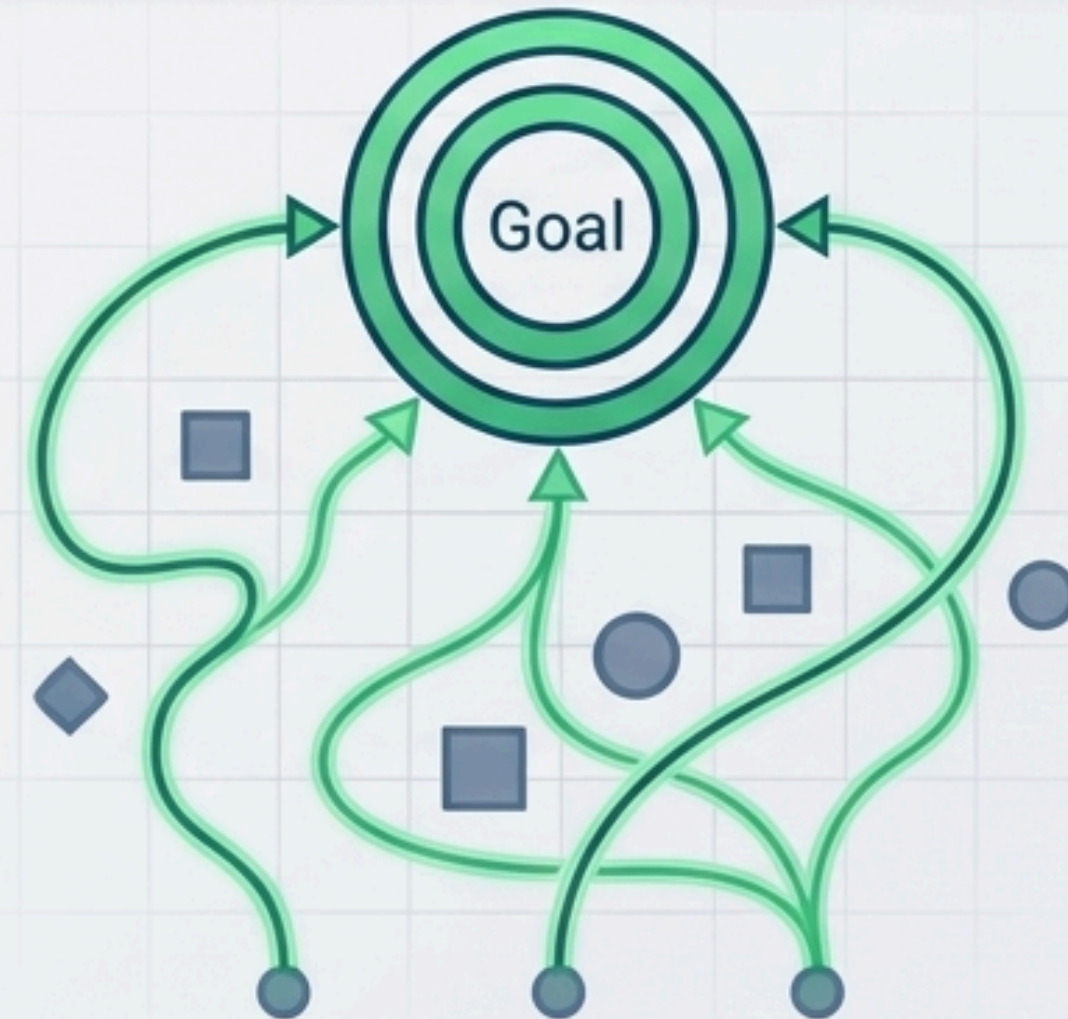
The How (Step-by-Step Procedure)



Do not dictate the step-by-step procedure.
Rigidly prescribed ladders are brittle.

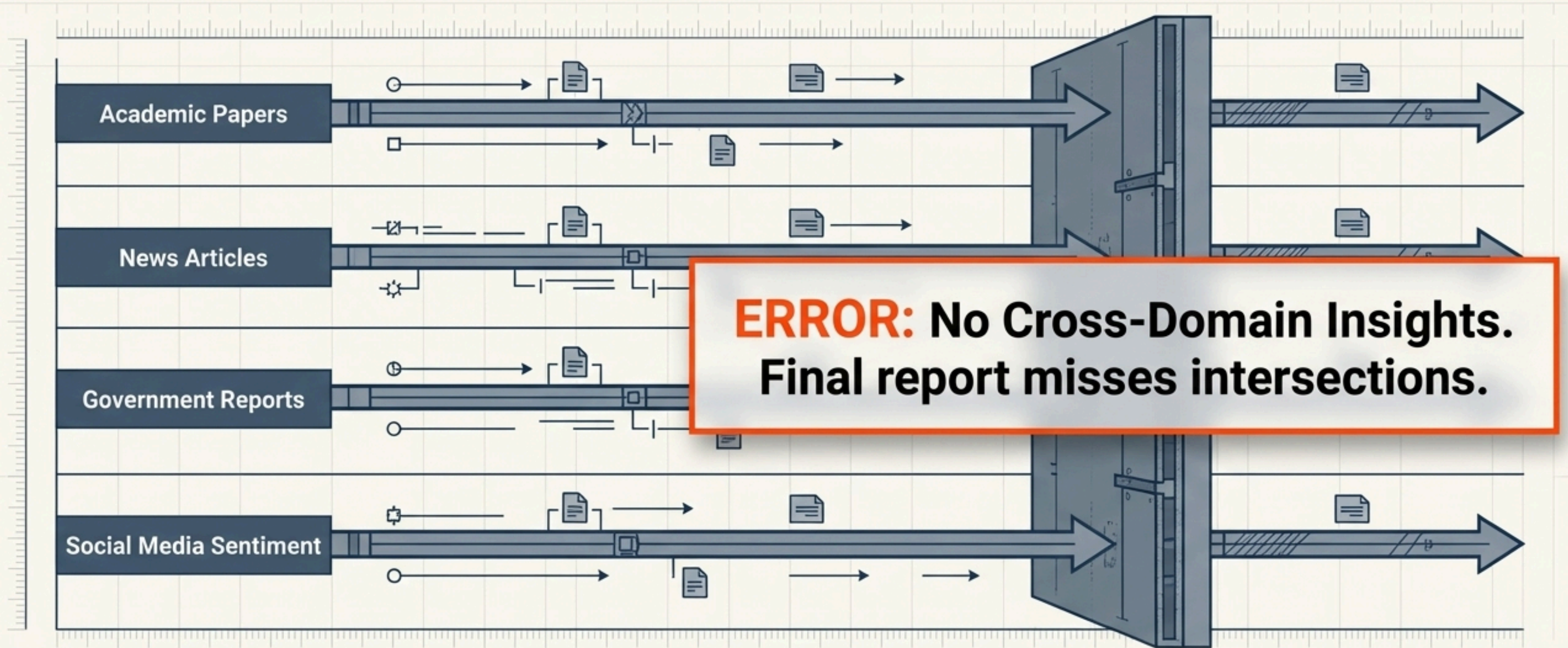
Best Practice

The What & Why (Goals + Quality Criteria)



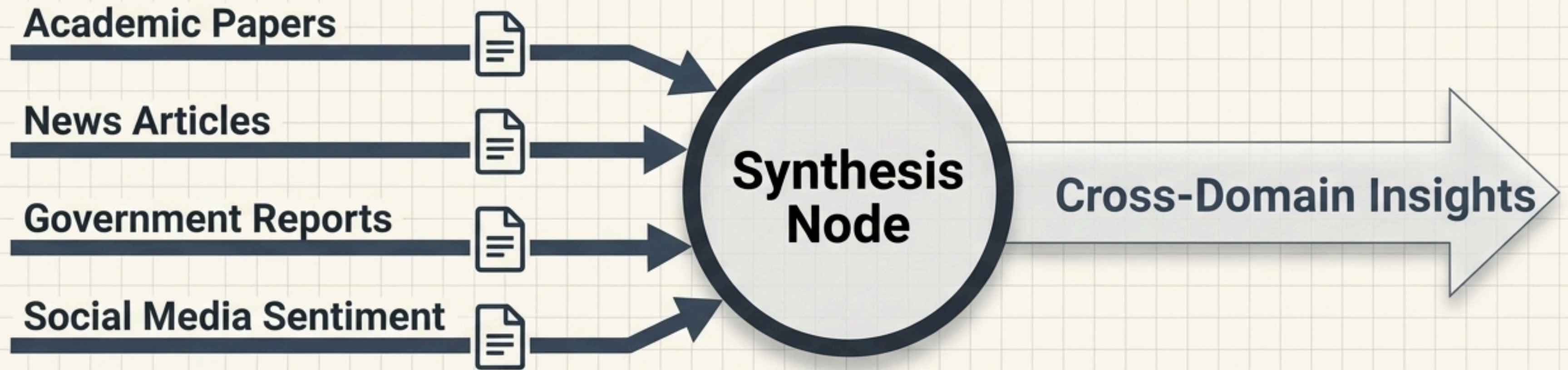
Outline the research goal and exact quality criteria.
Empower specialized subagents to determine their
own optimal approach using their allowed tools.

Exam Forensic Map: The Siloed Research Failure



- **The Scenario:** Coordinator spawns 4 parallel subagents for a multifaceted research report.
- **The Symptoms:** Output is fast, but economic data and policy timelines never intersect. Each agent stays perfectly in its lane.
- **The False Culprits:** It is NOT a small context window, wrong tools, or a bloated system prompt.

Fixing the Pipeline: The Synthesis Node



The Root Cause: Task decomposition was too narrow. Four independent lanes existed, but **no single entity was responsible for cross-domain synthesis**.

The Fix: Add an explicit **aggregation/synthesis step**. Spawn a 5th cross-domain subagent, or force a final coordinator pass that explicitly seeks intersections.

Exam Pattern: When a multi-agent system yields incomplete or siloed outputs, it is almost always a missing step in the pipeline, not a model configuration issue.

Domain 1 Architecture Cheat Sheet

Concept	The Hard Rule	Exam Diagnostic (Why it matters)
Topology	All communication routes through the coordinator.	Diagnostic: Enables absolute observability.
Task Tool	Coordinator needs it; Subagents don't.	Diagnostic: Prevents recursive delegation loops.
Parallel Spawning	Multiple Task calls in ONE response.	Diagnostic: Fixes 3x latency issues.
Context Isolation	Subagents inherit zero history.	Diagnostic: Always explicitly inject context.
Siloed Outputs	Usually caused by overly narrow decomposition.	Diagnostic: Pipeline is missing a synthesis/aggregation step.